

5.

Figure 5.1

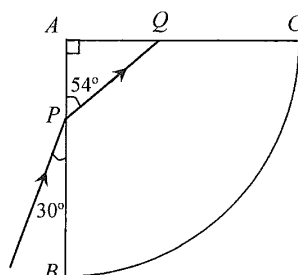


Figure 5.1 shows the cross-section of a glass block ABC . ABC is a quarter circle with its centre at A . A ray of red light is incident at P on face AB and the refracted ray strikes the face AC at Q as shown.

- (a) Calculate the refractive index of the glass for red light. (2 marks)

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- (b) Explain why the ray is totally reflected when it strikes the face AC at Q . (2 marks)

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- (c) In Figure 5.1, sketch the subsequent path of the ray until it finally emerges from the block to the air. (2 marks)

- (d) If the incident ray is a ray of white light, what can be observed when it finally emerges from the block? (1 mark)

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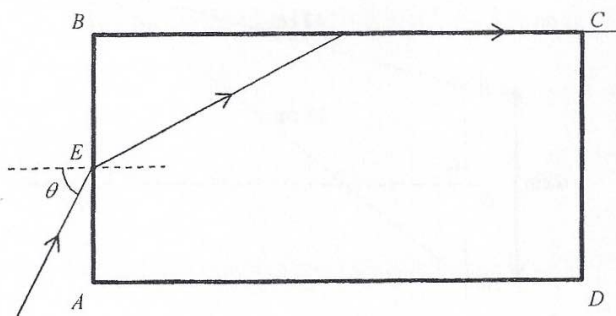
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Answers written in the margins will not be marked.

7. (a) A light ray enters a rectangular plastic block $ABCD$ from air at point E , and the angle of incidence is θ . The light ray emerges along face BC as shown in Figure 7.1. The refractive index of the plastic is 1.36.

Figure 7.1



- (i) Find the critical angle of the plastic. (2 marks)

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- (ii) Find the value of θ . (3 marks)

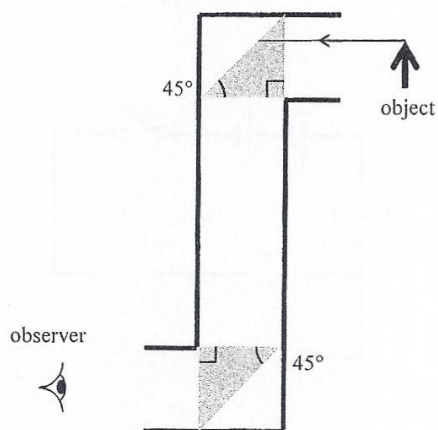
- (iii) If the light ray enters the plastic block at point E with an angle of incidence larger than θ , sketch the path of the light ray in Figure 7.1. (2 marks)

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- (b) A student designs a periscope using two plastic prisms, the refractive index of the plastic is 1.36. As shown in Figure 7.2, an object is placed in front of the periscope.

Figure 7.2



- (i) Complete the path of the light ray from the object in Figure 7.2, and explain why the periscope fails to work. (3 marks)

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- (ii) What can be used to replace the two plastic prisms so that the periscope can work properly? (1 mark)

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7. Figure 7.1 shows an optical fibre which consists of a cylindrical glass core of refractive index n_g enclosed by a transparent cladding of refractive index n_c .

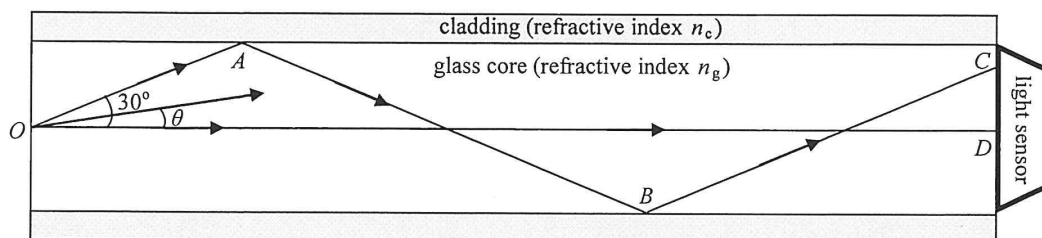


Figure 7.1

As shown in Figure 7.1, a point light source at O emits monochromatic light in all directions. Inside the fibre, light can reach the right end of the fibre through many different paths making angles θ with the axis OD . Two of these paths, OD and $OABC$, have been drawn for reference. Light ray OA makes an angle of 30° with the axis OD and is incident at the core-cladding boundary at A with an angle of incidence i_A .

- (a) (i) Find i_A . (1 mark)

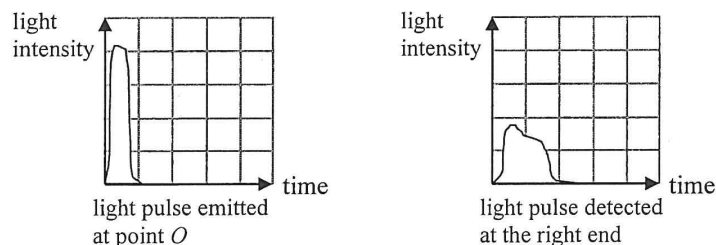
- (ii) If i_A is just greater than the critical angle of that boundary, estimate $\frac{n_g}{n_c}$. (2 marks)

- (iii) What phenomenon occurs at point A ? State the condition needs to be satisfied by θ such that this phenomenon **fails to occur**. (2 marks)

Answers written in the margins will not be marked.

- (b) A narrow monochromatic light pulse (i.e. of a short duration) emitted at point O propagates with its energy within $\theta = \pm 30^\circ$ towards a light sensor located at the right end of the optical fibre. The respective emitted and detected light pulses are represented below using the same scales.

Figure 7.2



- (i) Explain why the light pulse detected is **broad**er (i.e. of a longer duration) and with **lower intensity**. Assume that the loss of energy of the light pulse due to absorption by glass is negligible. (2 marks)

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- (ii) An engineer suggests changing the refractive index n_c of the cladding in order to reduce the width of the light pulse received. Should n_c be increased or decreased? Or, will a change in n_c have no effect on the pulse width? Explain your choice. (2 marks)

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6. Read the following passage about **rainbows** and answer the questions that follow.

Rainbows are sometimes seen after a rainfall (Figure 6.1(a)). Tiny water droplets in the air each act like a prism to disperse sunlight into a spectrum of colours. The most common rainbow that we see is called a primary rainbow, in which the white light from the sun undergoes reflection once within each water droplet.

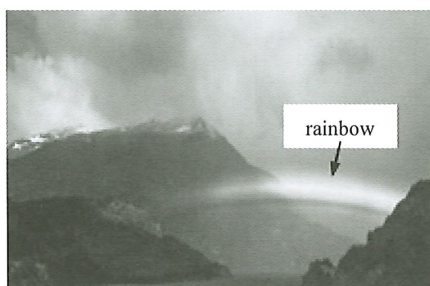


Figure 6.1(a)

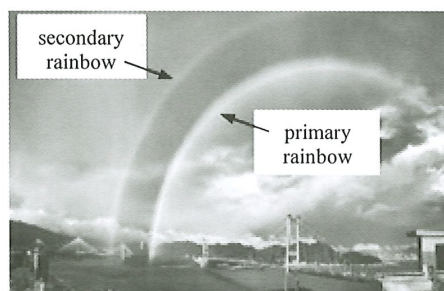


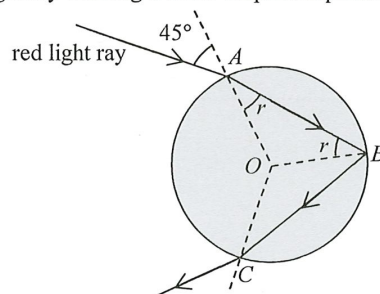
Figure 6.1(b)

Under suitable conditions, a double rainbow with an additional secondary rainbow formed above the primary rainbow can be seen (Figure 6.1(b)).

Assume that water droplets are spherical. Given: refractive index for red light in water = 1.325

- (a) Figure 6.2 shows a red light ray entering a water droplet suspended in air.

Figure 6.2

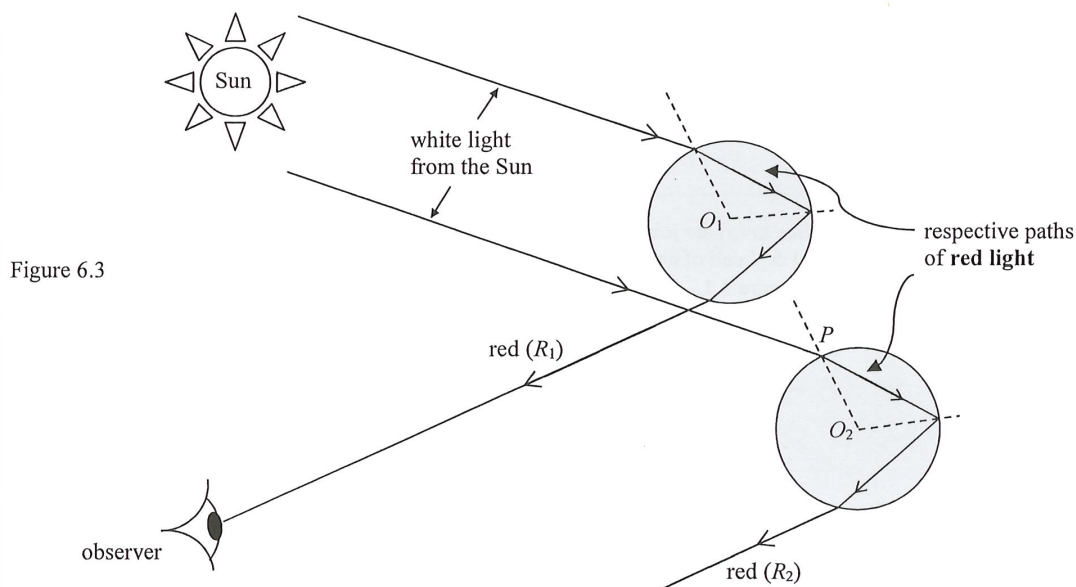


- (i) When the angle of incidence is 45° , find the angle of refraction r at point A . (2 marks)

- (ii) Calculate the critical angle c for red light at the water-air interface and justify whether the reflection at point B is a total internal reflection or not. (3 marks)

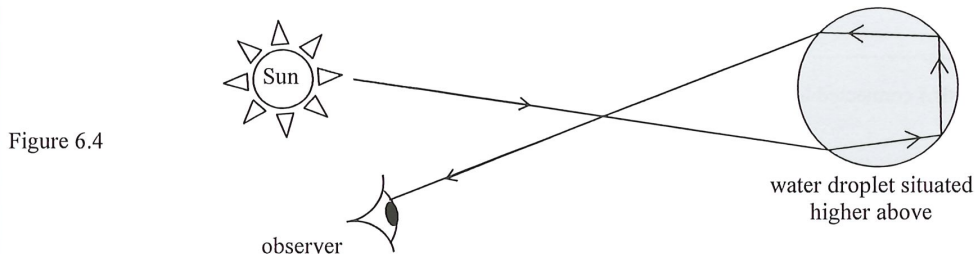
Answers written in the margins will not be marked.

- (b) (i) Figure 6.3 illustrates how an observer sees a rainbow. O_1 and O_2 are the respective centres of the upper and lower water droplets suspended in air. The two paths of red light, R_1 and R_2 , in the white light from the Sun are drawn for your reference, with R_1 being seen by the stationary observer.



The lower water droplet enables a violet light ray from sunlight to be seen by the observer. Sketch the path of this violet light ray starting from point P on the lower water droplet. Given that the refractive index for violet light in water is slightly greater than that for red light. (2 marks)

- (ii) For a secondary rainbow to be formed, a light ray has to undergo **reflection twice within each of those water droplets** situated higher above as shown in Figure 6.4.

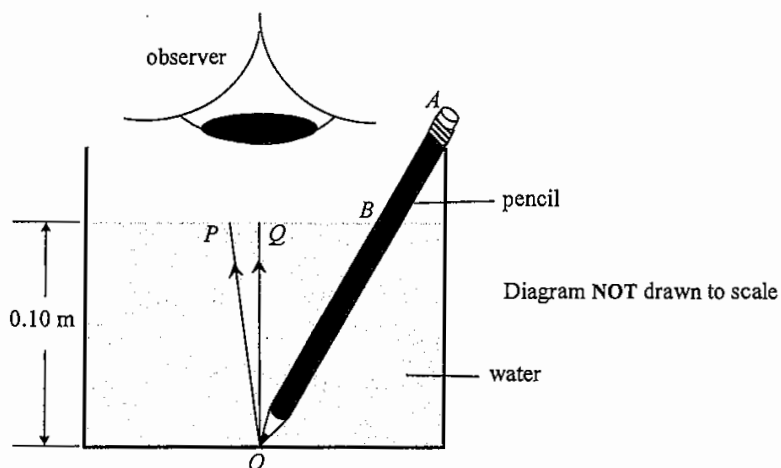


Explain why a secondary rainbow is dimmer than the primary rainbow. (2 marks)

Answers written in the margins will not be marked.

8. Figure 8.1 shows a pencil in a glass of water with an observer viewing vertically downwards. Given that the refractive index of water is 1.33.

Figure 8.1



Light rays P and Q come from the tip O of the pencil, with Q along the vertical.

- (a) On Figure 8.1, complete the paths of these light rays and indicate the image of O seen by the observer (label this as I). (2 marks)
- (b) The angle between ray P and ray Q is 1.5° . Find the angle of refraction of ray P . (2 marks)

- (c) Hence, or otherwise, find the distance between image I and the water surface if the depth of water is 0.10 m. (2 marks)

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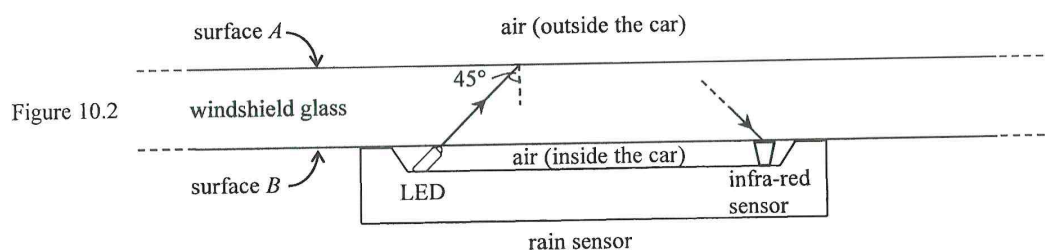
Answers written in the margins will not be marked.

10. Figure 10.1 shows a rain sensor mounted on the windshield glass of a car. It senses rainwater on the glass and automatically turns on the windshield wipers.

Figure 10.1



As shown in Figure 10.2, the rain sensor is composed of a light emitting diode (LED) and an infra-red sensor. The LED emits an infra-red signal and the infra-red sensor measures the intensity of the signal. When the intensity changes, the wipers are activated. An infra-red ray from the LED incident at surface A at 45° , and the infra-red ray finally received by the infra-red sensor are drawn in Figure 10.2. Assume surface A is parallel to surface B . It is given that the critical angle of the glass for infra-red is 44.0° .



- (a) Explain how the infra-red ray in Figure 10.2 travels from the LED to the infra-red sensor. (4 marks)

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Answers written in the margins will not be marked.

- (b) When it rains, some raindrops settle on the glass surface, as illustrated in Figure 10.3(a). Figure 10.3(b) shows an enlarged view of the region where the infra-red ray is incident on the glass-water interface.

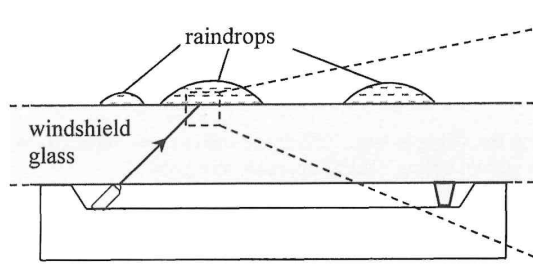


Figure 10.3(a)

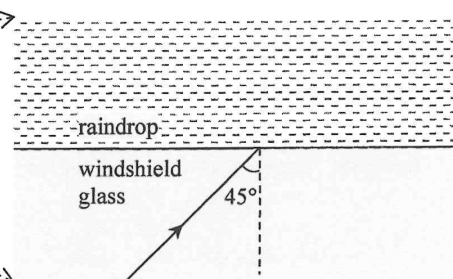


Figure 10.3(b)

- (i) Given that the refractive index of water for the infra-red is 1.26, find the critical angle of the glass-water interface for the infra-red ray. (3 marks)

- (ii) Sketch in Figure 10.3(b) how the infra-red ray propagates after being incident on the glass-water interface. (2 marks)

- (iii) The wipers are activated when the intensity of the signal received by the infra-red sensor changes. Explain briefly how the intensity changes. (2 marks)

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